

HOBART HUB

Technical Handover Document

Bar Wa Izakaya & Lost In Asia

Hobart, Tasmania

Project Overview

Hobart Hub is a single-page hospitality operations platform built for two venues: Bar Wa Izakaya and Lost In Asia in Hobart, Tasmania. It is a pure HTML/CSS/JavaScript application (no frameworks) with Firebase Firestore for cloud sync and localStorage for offline caching.

Tech Stack

- Frontend: Vanilla HTML + CSS + JavaScript (no build tools, no frameworks)
- Database: Firebase Firestore (cloud sync) + localStorage (offline cache)
- File Storage: Firebase Storage (invoices, rosters, documents, photos)
- AI: Google Gemini 2.5 Flash API (recipe parsing, invoice extraction, allergen detection, run sheets)
- Integrations: Tanda API (roster/timesheets/labour), Lightspeed POS (CSV imports), Open-Meteo (weather)
- PWA: Service worker + manifest.json for offline iPad support

File Structure

All source files live in: hubwa-source/hubwa-main/

- index.html (230 lines) — App shell, sidebar nav, Firebase init, section state management
- core.js (~1,450 lines) — Global state, PIN system, modals, Firebase sync, Tanda integration, search, notifications, venue management
- ops.js (~4,650 lines) — Inventory, recipes, ordering, invoices, wastage, compliance data ops, stocktake, stock audit, AI features
- mgmt.js (~3,700 lines) — Dashboard, analytics, staff hub, maintenance, handover, incidents, roster, digital safe, knowledge base
- style.css (~590 lines) — Dark theme, responsive design, component styles
- sw.js — Service worker for offline caching
- manifest.json — PWA manifest

Architecture

Data Flow

All data lives in global window arrays (e.g. `window.inventoryItems`, `window.recipes`). The `saveToDisk()` function in `core.js` serializes all arrays to both `localStorage` (immediate) and `Firebase Firestore` (debounced 3s). On startup, data loads from `localStorage` first (instant render), then `Firebase` syncs in the background.

Save Keys (41 data arrays)

`inventoryItems`, `recipes`, `wastageLogs`, `suppliers`, `salesData`, `salesTargets`, `orientationLogs`, `rotationalTasks`, `taskHistory`, `tempLogs`, `complianceLogs`, `defectLogs`, `equipmentData`, `contractorLogs`, `digitalSafe`, `phoneBook`, `incidentLogs`, `handoverLogs`, `knowledgeBase`, `shiftRosters`, `onboardingTemplates`, `fridgeUnits`, `masterChecklists`, `posMappings`, `storageZones`, `depletionLogs`, `safeCategories`, `kbCategories`, `orderHistory`, `staffDirectory`, `IsImportLog`, `IsSalesByData`, `shiftChecklistItems`, `invoiceMatchMap`, `priceHistory`, `inventorySubcategories`, `kbSubcategories`, `safeSubcategories`, `handoverTemplateConfig`, `qualificationTypes`, `stockMovements`, `stocktakes`

View Router

`window.showView(viewName)` in `core.js` is the master router. It maps string view names to render functions. Views return HTML strings that get injected into `#mainContent` via `innerHTML`. Hub wrappers (`renderInventoryHub`, `renderRecipeHub`, `renderAnalyticsHub`) use tab state variables and delegate to sub-view render functions.

Multi-Venue

Two venues defined in `window._venues` array. Data isolated by venue ID prefix in `localStorage` keys and separate `Firebase` documents (`hobartHub` and `lia`). `getCurrentVenue()` checks URL params, then `localStorage` `hubActiveVenue`, then `hubDeviceVenue`, then defaults to `bwi`.

Navigation Structure

Sidebar has 7 collapsible sections with 14 nav items. Many features are accessed via tab bars within parent views:

Sidebar Sections

- Daily Ops: Inventory Hub, Wastage Tracker, Compliance
- Ordering: Order Hub (3 tabs: Order List, AI Suggester, Invoice Ripper)
- Recipes & Costing: Recipe & Costing (5 tabs: Recipes, Margins, Ingredients, Allergens, Run Sheet)
- Analytics: Financials (4 tabs: Takings, Prime Cost, Variance, Forecast)
- Team & Venue: Staff Hub (4 tabs), Rotational Tasks, Maintenance, Handover, Incidents, Roster, Digital Safe, Knowledge Base
- Settings: Suppliers, Storage Zones, Tanda Settings
- External Links: Tanda, Lightspeed, Me&u

Tab Bar Pattern

Reusable pattern: state variable (`window._xxxTab`), array of `{id, label}` objects, `.tag-pill` elements with active class, onclick sets state and calls `showView()` on parent. Used by: Inventory Hub, Recipe Hub, Analytics Hub, Order Hub, Staff Hub, Compliance.

Security Implementation

PIN System

Forced setup on first load if no PIN in localStorage. Styled numeric keypad (3x4 grid) with dot indicators. PIN stored as plaintext in localStorage (venuePin). Auto-lock after 10 min inactivity.

Three-Tier Confirmations

- Standard: styled modal with confirm/cancel (routine actions)
- Dangerous: PIN required before confirm (bulk delete, stock reset)
- Critical: PIN + type confirmation word (venue data wipe)

Restricted Views

Binary lock/unlock (no role-based access). When locked, .restricted class nav items hidden and restricted views redirect to dashboard. Restricted: sales, suppliers, recipes, invoice, staff, safe, handover, margins, forecast, prime-cost, batch-linker, etc.

XSS Prevention

- window.esc() — HTML entity escaping for all user content in innerHTML
- window.safeUrl() — blocks javascript:, data:text/html, vbscript: protocols
- window.escAttr() — escapes quotes/backslashes for onclick attributes
- YouTube-only iframe embeds — non-YouTube URLs rendered as safe links

Integration Details

Tanda API

OAuth token stored in localStorage (tandaToken). All calls via window.fetchTanda(endpoint) wrapper with Authorization header. Data pulled: users, schedules (weekly roster), timesheets, clocked-in staff, leave requests. Auto-refresh: full every 15 min, clocked-in every 5 min. Merges into staffDirectory. Labour cost calculated from timesheets.

Lightspeed POS

Manual CSV upload only (no live API). Three CSV types: Sales By (product sales for depletion), Guests (covers data), Reconciliation (EOD totals). AI POS Depletion: paste product mix text, Gemini extracts items, matches to recipes via posAlias, cascades through ingredients including batch sub-recipes.

Google Gemini 2.5 Flash

API key stored in localStorage. Used for: AI recipe import, batch ingredient linker, allergen scanning, invoice text/vision extraction, POS depletion parsing, run sheet generation, order suggestions. All calls wrapped with loading overlay. Error handling with toast notifications.

Firebase

Firestore: venueData collection with docs per venue. Backups collection for daily snapshots (7-day retention). Storage: files uploaded to rosters/, knowledge/, safe/ prefixes. Write retry: 3 attempts with 10s backoff, clickable retry link on failure. Sync status badge in header.

Development History

Phase 1-3 (Pre-existing)

Core platform built: inventory management, recipe engine, compliance, staff management, Firebase sync, Lightspeed CSV import, Invoice Ripper, wastage tracking, digital safe, knowledge base, rotational tasks, maintenance, handover, incidents.

Phase 4: Navigation Restructure

- Sidebar reorganised into 6 collapsible sections
- Staff Hub: merged Directory + Onboarding + Qualifications + Phonebook
- Compliance: merged Temps + Shift Checklists + Custom + HACCP
- Order Hub: added tab bar for Order List + AI Suggester
- Parent view button pattern for easy sub-navigation

Batch 1: Bug Fixes & Qualifications

- Replaced all 21 confirm() dialogs with styled 3-tier modal system
- Added autofocus on all modal inputs
- Debounced inventory search (250ms)
- Fixed --text-muted contrast for WCAG AA
- Added empty string validation to key forms
- Built editable staff qualification tracking system
- Forced PIN setup on first load with styled PIN pad

Batch 2: Dashboard & Security

- Complete dashboard rewrite (Venue Pulse) with health score, financial cards, charts
- URL sanitization (safeUrl) on Digital Safe links and roster embeds
- onclick injection prevention (escAttr) on onboarding and checklist templates

Tier 1: Critical Bug Fixes

- Replaced last prompt() calls with styled modals (inline stock edit, PAR edit)
- Yield validation: min 0.01, prevents divide-by-zero
- Dangling recipe refs: unlinks ingredients when inventory items deleted
- Loading overlay on all 8 Gemini AI calls
- Bulk archive confirmation added
- Recipe video embeds sanitized (YouTube-only iframes)
- Modal body scroll lock for Safari iPad
- Firebase retry logic: 3 retries, 10s backoff, clickable retry on failure

Tier 2: UX Polish

- Toast stacking (vertical, not overlapping)
- Mobile sidebar auto-close on nav tap
- Keyboard shortcuts: Cmd/Ctrl+K (search), Cmd/Ctrl+S (save), Esc (close modal)
- Search result text highlighting
- Inventory CSV export
- Recipe cost auto-recalculation on ingredient price changes
- Stocktake workflow
- Dark/light mode toggle

Tanda Deep Integration

- Full week roster pull (not just today)
- Clocked-in staff with resolved names
- FOH/BOH department grouping on roster view
- Labour cost calculation from timesheet data

Nav Consolidation: 25 !' 14 Items

- Created 4 hub wrappers: Inventory Hub, Recipe Hub, Analytics Hub, Order Hub
- Absorbed 11 nav items into parent tab bars
- Backward-compatible routing (old view names still work)

Security Sweep & Polish

- XSS patch in global search onclick handlers
- localStorage quota guard (warns at 80% capacity)
- Replaced last 3 prompt() calls (API key, recipe scale, roster name)
- Global search debounced (250ms)
- Console.log cleanup for production

Tier 3: Architecture & Scale

- PWA: manifest.json + service worker for offline iPad support
- Performance: 200-item render cap on inventory view
- Auto-backup: daily Firebase snapshots, 7-day retention
- Storage monitor: warns at 4MB usage
- Log trimming: stockMovements(500), depletionLogs(200), wastageLogs(200)

Remaining Work & Known Issues

High Priority

- Live Stock Depletion Pipeline: unify CSV import and AI depletion paths so both cascade through recipes identically
 - Role-Based Access: currently binary lock/unlock — needs manager vs staff roles with granular view permissions
 - Offline Data Conflict Resolution: if user edits data offline and Firebase has newer data, no merge strategy exists
 - PIN stored as plaintext in localStorage — should be hashed

Medium Priority

- Lightspeed live API integration (currently CSV-only, manual upload)
- Multi-venue shared resources (recipes, suppliers) — currently fully isolated
- IndexedDB migration for large datasets (localStorage 5MB limit)
- Automated reporting: scheduled email summaries
- Print-friendly PDF exports for compliance, inventory, recipes
- Cross-venue dashboard: aggregate KPIs across both venues

Low Priority / Nice to Have

- Accessibility: nav divs should be semantic buttons with ARIA labels
- Service worker cache versioning: bump CACHE_NAME on code changes
- Recipe photo upload to Firebase Storage (currently URL-only)
- Historical analytics: month-over-month trending, year-on-year comparison
- Staff scheduling: build rosters directly in Hub instead of Tanda
- Customer/booking management integration (SevenRooms API)

Key Code Patterns

View Rendering

All views are functions that return HTML strings. `window.showView(name)` injects the result into `#mainContent`. Hub wrappers check a tab state variable and delegate to sub-view functions.

Modal System

`window.openModal(title, bodyHtml)` renders into `#global-modal-content` overlay. Patched with autofocus and body scroll lock. `window.confirmAction({title, message, tier, onConfirm})` for confirmations.

Toasts

`window.showToast(msg, type)` creates floating notifications. Types: default (green), error (red). Auto-dismiss after 3s. Stacks vertically.

Firebase Sync

`window.saveToDisk()` writes to `localStorage` immediately, then debounces Firebase write by 3s. Retry logic on failure. Daily backup on first successful sync each day.

Data Escaping

`window.esc(str)` for innerHTML content. `window.escAttr(str)` for onclick attributes. `window.safeUrl(url)` for href/src attributes. Always use these when injecting user data.

Owner Preferences

- Prefers batch changes over incremental
- Discuss approach before coding
- Short responses preferred
- Everything should be editable by the user (avoid hardcoding)
- iPad is the primary device — touch targets and scroll behaviour matter
- Two venues: Bar Wa Izakaya (primary) and Lost In Asia